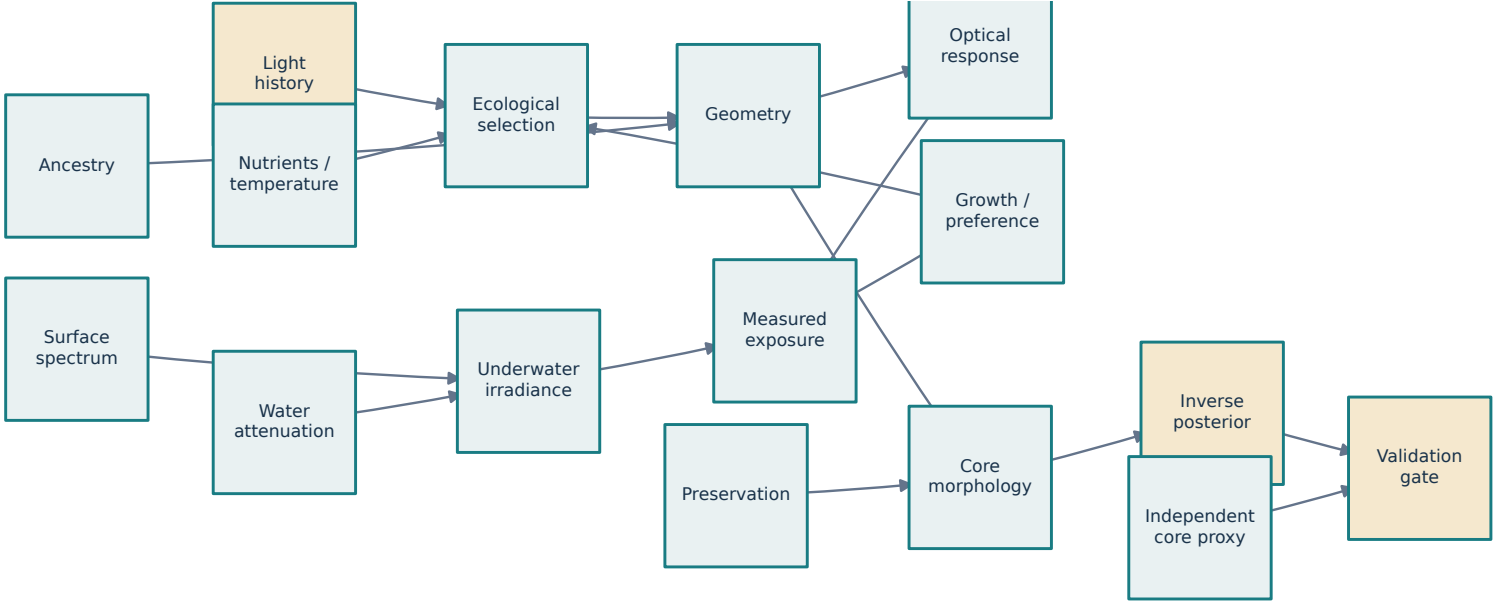Causal dependencies: optical forward physics is only one layer